

Christopher Williams

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EDUCATION

MEng. Computer Science

May 2026

Virginia Tech, Alexandria, VA (*Data Viz & AI*)

Coursework: Machine Learning, NLP, Database Systems, Urban Computing

B.S. Computer Science

May 2025

Christopher Newport University, Newport News, VA

Coursework: Artificial Intelligence, Quantum Mechanics, Communications Networks, Algorithms & Data Structures

SKILLS

ML Frameworks: PyTorch (DDP multi-GPU, open-source contributor), Stim, SciPy, scikit-learn

Distributed & HPC: PyTorch DDP, SLURM/HPC clusters (A100/H200 GPUs), HDF5, Docker

Languages: Python, C++, Java, JavaScript/React, Bash, LaTeX

Systems & Tools: Linux, Git, Docker, ROS, NetworkX, Pandas

Domains: ML/AI Systems, Quantum Error Correction, Transformer Architectures, Robotics

EXPERIENCE

Student Researcher — Christopher Newport University

June 2023 – July 2023

Newport News, VA

- Developed autonomous mobile robot navigation using pedestrian ML pathfinding; integrated **YOLOv8** + **OpenCV** via **ROS** in **C++/Python**

PROJECTS

Quantum Error Correction Transformer Decoder

Sept. 2025 – Present

Python, Stim, PyTorch (DDP), Docker, HDF5, Bash, HPC's — *Virginia Tech & Northrop Grumman*

- Building a transformer-based QEC decoder grounded in Google DeepMind's AlphaQubit, extending decoding capabilities beyond surface code to advance fault-tolerant quantum computing, in active collaboration with Northrop Grumman engineers Michael Mullan and Dusty Johnson
- Designed distributed training pipeline across **4 H200 GPUs** via **PyTorch DDP** and **SLURM**; engineered **HDF5** compression pipeline for **50.5GB** of syndrome data generated with **Stim** and sourced from Google, containerized with **Docker**
- Achieved **0.25% improvement** over MWPM baseline LER for distance-3 surface code in prior decoder iteration

Population Flow Prediction via Exchange Rates

Nov. 2025 – Present

Python, NetworkX, PyTorch, Pandas — *Virginia Tech*

- Predicting large-scale population flows through currency exchange rate analysis using publicly available datasets in cross-domain collaboration with Dr. Naren Ramakrishnan
- Analysis done in **Python**, utilizing the Amazon Chronos **PyTorch** time series model and **NetworkX** visualizations to model bilateral migration patterns at international scale

Sarcasm Detection with NLP

Jan. – May 2026

Python, Transformers, Bash — *Virginia Tech*

- Fine-tuned **encoder transformers** (BERT, RoBERTa, DeBERTa, ELECTRA, DistilBERT) on public datasets to classify sarcasm in text, with applications to sentiment analysis accuracy in social media content moderation

Steam Game Recommender

Jan. – May 2026

FastAPI, React, MongoDB — *Virginia Tech*

- Natural language game recommender that uses **React** + **Vite** frontend connected to a **MongoDB** backend and queries sent to an LLM for recommendations, with recommendations resolved against a curated MongoDB game catalog

Gentrification Prediction in the DMV with Machine Learning

Sept. – Dec. 2025

Python, XGBoost, NetworkX, Go, SQL, Svelte, Docker — *Virginia Tech*

- Built a 5-year county-level gentrification forecasting tool for the DMV to support urban policy analysis and housing planning decisions; trained an **XGBoost** model on public census data with full-stack pipeline using **Go/SQL** backend, **Svelte** frontend, and **NetworkX** visualizations, containerized with **Docker**